Coverage-Aware Selection for Vision-Language Instruction Tuning

Anonymous ACL submission

Abstract

Vision-language instruction tuning relies on large heterogeneous mixtures, but a data-centric paper must separate genuine data efficiency from artifacts of backbone choice, benchmark choice, and prompt handling. This draft studies a coverage-aware quality selection strategy for LLaVA-1.5-7B LoRA tuning that keeps the training recipe and evaluation harness fixed while varying only the instruction-data subset. The planned evaluation compares the full-data tuning anchor against no-additional-tuning, random, scalar-quality, diversity-only, ShareGPT4V-style when accessible, and coverage-aware subsets on MMBench, MME, SEED-Bench, and TextVQA. The central result is intentionally unresolved: `[TBD]`, with benchmark-specific cells `[TBD]`, `[TBD]`, `[TBD]`, `[TBD]`, `[TBD]`, and `[TBD]`. All significance, failure-case, and outcome statements remain placeholders until canonical scored rows exist: `[TBD]` and `[TBD]`.

1 Introduction

Vision-language instruction tuning has become a practical way to adapt multimodal models to user-facing visual reasoning, question answering, and text-in-image tasks, building on image-text pretraining, frozen-model VLM adaptation, and instruction-tuned multimodal assistants (Radford et al., 2021; Alayrac et al., 2022; Li et al., 2023c; Liu et al., 2023b; Dai et al., 2023; Zhu et al., 2023; Ye et al., 2023). The cost of this adaptation is not only the backbone model; it is also the size, provenance, and mixture balance of the instruction data. The project therefore asks a narrow data-centric question: under a fixed LLaVA-1.5-7B LoRA recipe, can a subset selected for coverage over source, answer regime, and failure-mode signals preserve benchmark-relevant behavior with less data than full-data tuning (Liu et al., 2023a)?

The motivation is grounded in the project literature survey rather than in completed results. The local research brief identifies visual instruction tuning, stronger caption-style data, large multimodal instruction mixtures, and current open VLM training stacks as relevant evidence families (Li et al., 2023d,a; Zhao et al., 2023; Chen et al., 2023b; Xu et al., 2024; Tong et al., 2024; Farina et al., 2026). It also identifies MMBench, MME, SEED-Bench, and TextVQA as the required benchmark families for the paper-facing comparison (Liu et al., 2023d; Fu et al., 2023; Li et al., 2023b; Singh et al., 2019). This manuscript scaffold uses those verified planning facts only; it does not claim that the proposed selector wins, matches, or improves any benchmark.

The proposed selector, Coverage-Aware Quality Selection (CAQS), treats data quality as a gate rather than a single ranking objective, following alignment-data findings that small or automatically selected instruction sets can be useful only when selection criteria are explicit and auditable (Zhou et al., 2023; Liu et al., 2023c). After invalid records and unusable image-text pairs are filtered, the selector balances coverage over source provenance, answer regimes, and benchmark-relevant failure-mode tags while penalizing near-duplicate examples. The hypothesis is that a small subset should not merely contain fluent examples; it should retain complementary visual and answer behaviors that are easy to lose under scalar quality ranking.

The first experimental dependency is the full-data baseline, not the curated subset. Without that anchor, a subset result could be misread as data efficiency when it actually reflects a weak recipe, an evaluator mismatch, or an uncollected training failure. The current draft therefore treats the full-data run as an unresolved prerequisite, represented throughout as `[TBD]`. This keeps the manuscript useful for overlap drafting while preventing the abstract, results, and conclusion from implying that the data-centric thesis has already been tested.

This paper will make three contributions once the evidence exists. First, it will provide a controlled LLaVA-1.5-7B LoRA comparison in which the data subset changes while the backbone, training framework, and evaluation harness are held fixed. Second, it will compare CAQS with full data and same-size controls, including random, scalar-quality, and diversity-only subsets. Third, it will analyze whether the selector's behavior is explained by coverage signals rather than by data fraction alone. The current result status is `[TBD]` and every outcome sentence in the draft remains a placeholder.

2 Related Work

Visual instruction tuning. LLaVA-style instruction tuning made assistant-style multimodal supervision a practical baseline, and LLaVA-1.5 provides the fixed full-data recipe used as this project's training anchor (Liu et al., 2023b,a). InstructBLIP, MiniGPT-4, mPLUG-Owl, OpenFlamingo, Qwen-VL, and Shikra study adjacent ways to connect language models with visual inputs, prompt following, text reading, localization, or referential interaction (Dai et al., 2023; Zhu et al., 2023; Ye et al., 2023; Awadalla et al., 2023; Bai et al., 2023; Chen et al., 2023a). Later open systems such as Cambrian-1 and LLaVA-OneVision emphasize that multimodal papers now need to expose data, recipes, checkpoints, and evaluation procedures rather than only final scores (Tong et al., 2024; Li et al., 2024; An et al., 2025, 2026).

Data quality, synthesis, and mixture construction. Large multimodal instruction mixtures and synthetic instruction data motivate broad source coverage, while caption-focused data improvements motivate quality-aware construction (Li et al., 2023d,a; Zhao et al., 2023; Chen et al., 2023b; Xu et al., 2024). DataComp-VLM further frames open dataset construction as an active VLM research object rather than a hidden preprocessing choice (Farina et al., 2026). The paper's gap is not whether better captions or larger mixtures can help in general; it is whether a coverage-aware subset can be compared fairly against scalar quality and diversity controls under the same LLaVA-1.5 recipe.

Data selection and coverage. Instruction-tuning data selection work argues that example quality, diversity, and task fit can matter as much as raw volume, but these signals are easy to conflate when the model, training budget, and evaluation suite also change (Zhou et al., 2023; Liu et al., 2023c). The planned method separates four roles that are often conflated: a quality gate removes broken examples, answer-regime tags preserve response-format diversity, source balancing prevents one dataset family from dominating, and redundancy penalties avoid many polished but equivalent rows.

Evaluation confounds in data-centric VLM work. A data-selection claim is easy to overstate when one condition uses a different checkpoint, prompt template, decoding policy, or benchmark wrapper from another. This paper's design follows the stricter interpretation of data-centric evidence: the model family, adaptation recipe, evaluation harness, and benchmark splits should remain fixed while the data-selection rule changes. The remaining uncertainty should be visible in a run matrix, not absorbed into a single aggregate score. Until that matrix is executed, every comparison among full data, random, quality-only, diversity-only, and CAQS conditions remains `[TBD]`.

Multimodal evaluation. The required benchmark families cover broad multimodal reasoning, perception/cognition scoring, generative visual comprehension, and OCR-oriented visual question answering (Liu et al., 2023d; Fu et al., 2023; Li et al., 2023b; Singh et al., 2019). They sit in a broader evaluation lineage that includes visual question answering, knowledge-aware VQA, accessibility-oriented VQA, science-question reasoning, and integrated multimodal capability testing (Goyal et al., 2016; Marino et al., 2019; Gurari et al., 2018; Lu et al., 2022; Yu et al., 2023). The benchmark-facing paper claim will be limited to the four required task families unless later evaluation artifacts add more public benchmark evidence. The evaluation harness choice is also part of the paper contract: LMMs-Eval is cited as the maintained multimodal benchmark harness, while vLLM is relevant only as an inference-throughput reference rather than a scoring substitute (Zhang et al., 2024; Kwon et al., 2023).

3 Method

CAQS operates on the normalized LLaVA-1.5 instruction mixture (Liu et al., 2023a). Each candidate example is represented by source metadata, conversation structure, image availability, answer-regime features, and lightweight failure-mode tags.

The selector first applies a quality gate for schema validity, image availability, assistant-answer completeness, and duplicate checks. It then chooses examples using a coverage objective that rewards underrepresented regimes and penalizes redundant examples, keeping the quality and coverage roles separate in the spirit of prior instruction-data selection studies (Zhou et al., 2023; Liu et al., 2023c).

The selector is designed for a controlled data-centric comparison, following the broader expectation that VLM data papers report the data intervention separately from model and benchmark changes (Farina et al., 2026; Li et al., 2024). It does not change the model family, training objective, optimizer schedule, multimodal adapter recipe, or benchmark prompts between conditions. The full-data condition anchors the comparison; proposed subsets and controls are trained and evaluated under the same paper-facing recipe.

Selection units. The selection unit is a normalized instruction example with one visual input, one or more conversation turns, and a paper-facing source label. The normalized manifest records whether the image exists, whether the assistant answer is usable, whether the example resembles an OCR, yes/no, multiple-choice-like, captioning, numeric, or free-form answer regime, and which coarse visual skill tags are triggered. The exact manifest profile and retained counts are `[TBD]`. This placeholder is deliberately separate from result placeholders because a subset can be well formed yet still fail to improve benchmark behavior.

Quality gate. The gate removes examples that cannot be evaluated as valid instruction-tuning data: missing media, malformed conversations, empty assistant messages, duplicated records, and examples whose image/text pairing cannot be trusted from local provenance. The gate is not intended to score helpfulness or creativity. Its role is to keep broken rows from dominating later coverage decisions. The final paper must report how many rows the gate removes, but that profile is `[TBD]` until the selector manifest is frozen.

Coverage objective. After gating, CAQS treats source, answer regime, and failure-mode tags as coverage strata. It prefers examples that add underrepresented strata and downweights examples that are near-duplicates of already selected rows. A scalar-quality selector can rank many polished caption-like rows highly; CAQS instead asks whether the selected set still spans OCR, counting, recognition, spatial, and reasoning regimes that the benchmark suite will later probe. The final ablation must show whether these strata matter: `[TBD]`, `[TBD]`, and `[TBD]`.

Component	Paper-facing role
Quality gate	Removes unusable records before selection.
Source coverage	Keeps the subset from collapsing to one data source.
Answer-regime coverage	Preserves yes/no, multiple-choice-like, OCR, numeric, captioning, and free-form response patterns.
Failure-mode tags	Tracks visual skills such as text reading, counting, spatial relations, recognition, and commonsense.
Redundancy penalty	Discourages selecting many semantically similar examples.

Table 1: CAQS components planned for the controlled subset comparison; numerical effects are `[TBD]`.

Decision point	Evidence required before replacement
Full-data anchor	`[TBD]`; `[TBD]`; `[TBD]`
Subset manifest	`[TBD]`; `[TBD]`; `[TBD]`
Control fairness	`[TBD]` under the same recipe, split, and harness
CAQS comparison	`[TBD]` plus `[TBD]`
Ablation support	`[TBD]`; `[TBD]`; `[TBD]`; `[TBD]`

Table 2: Evidence gates for replacing placeholders with paper-facing claims.

4 Experimental Setup

Backbone and training recipe. The planned training anchor is LLaVA-1.5-7B with LoRA adaptation (Liu et al., 2023a). Training is routed through LLaMA-Factory so the paper can bind the data-centric comparison to a maintained fine-tuning framework rather than a self-written training loop (Zheng et al., 2024). The frozen full-data baseline uses one epoch over the measured LLaVA-style instruction manifest, a global batch size of 128, LoRA rank 128, LoRA alpha 256, cosine scheduling, and a maximum sequence length of 2048. These are plan and manifest facts, not result claims.

Conditions. The run matrix includes the full-data LoRA baseline, the no-additional-tuning base checkpoint, random same-size subsets, scalar quality-ranked subsets, diversity-only subsets, a

ShareGPT4V-style condition when data access permits, CAQS subsets, and CAQS ablations (Chen et al., 2023b). The primary subset budget is planned at 20 percent, with 10 percent and 30 percent scale checks.

Run order. The paper-facing run order is conservative. The full-data baseline must first produce a valid adapter inventory and training telemetry: [TBD] and [TBD]. Only then should subset and control conditions be launched, because the subset claim is meaningful only relative to a functioning full-data anchor. After each training condition, evaluation uses the same benchmark harness, model-loading policy, decoding budget, and answer-extraction policy unless the experiment plan records a justified exception.

Benchmarks and metrics. The required benchmark families are MMBench, MME, SEED-Bench, and TextVQA, executed through a maintained multimodal evaluation harness with benchmark-specific official or harness-supported metrics (Liu et al., 2023d; Fu et al., 2023; Li et al., 2023b; Singh et al., 2019; Zhang et al., 2024). The final table must report task split, evaluated model/backend, condition, metric, budget or decoding policy, and result. Those cells remain placeholders until raw scored rows exist.

Result collection contract. Each benchmark-condition row must be backed by raw scored rows, an aggregate artifact, and a claim-graph entry. A status file or planned task count is not sufficient evidence. The placeholder [TBD] is therefore attached to every benchmark row in the scaffold, and it can be removed only after the row has a canonical source in the result table.

5 Results

The result section is scaffolded before evidence exists. It does not report wins, losses, averages, or benchmark deltas. Table 3 reserves the paper's central evidence matrix and uses only placeholder tokens tracked by the scaffold registry.

The headline data-efficiency statement is [TBD]. Any claim that CAQS matches, beats, improves over, or fails relative to the full-data baseline is also [TBD] until generated from the canonical result table.

Full-data anchor. The first result paragraph in the final paper must answer whether the full-data LLaVA-1.5-7B LoRA baseline trained and evaluated correctly. Its evidence sources will include adapter inventory, training telemetry, and four benchmark aggregates. In this scaffold, that paragraph is limited to [TBD], [TBD], [TBD], [TBD], [TBD], [TBD], and [TBD].

Subset comparison. The second result paragraph should compare CAQS against the same-size controls and the full-data anchor. It must name the subset data fraction only after the subset manifests are frozen and must avoid broad claims when a benchmark family disagrees. The current placeholder bundle is [TBD], [TBD], [TBD], and [TBD].

Benchmark	Split	Registered result placeholders
MMBench	[TBD]	[TBD]; [TBD]; [TBD]; [TBD]
MME	[TBD]	[TBD]; [TBD]; [TBD]; [TBD]
SEED-Bench	[TBD]	[TBD]; [TBD]; [TBD]; [TBD]
TextVQA	[TBD]	[TBD]; [TBD]; [TBD]; [TBD]

Table 3: Planned cross-benchmark result matrix. All metric cells are placeholders until canonical scored rows and aggregate artifacts exist.

Ablation question	Placeholder evidence
Does source coverage matter?	[TBD]
Does answer-regime coverage matter?	[TBD]
Does redundancy control matter?	[TBD]
Are differences paired-stable?	[TBD]

Table 4: Planned ablation table; every cell remains unresolved until raw paired rows and aggregate tests exist.

6 Analysis/Ablation

The analysis will test whether coverage signals explain any observed behavior. Planned ablations remove or isolate answer-regime balancing, source coverage, failure-mode tags, scalar quality filtering, and redundancy penalties. The paired-comparison and uncertainty analysis is [TBD]. No ablation outcome is asserted in this scaffold.

The final analysis should also check whether a selector that improves one benchmark family degrades another. This matters because the proposed mechanism is coverage, not scalar maximization. A strong final claim would require a cross-family pattern and paired support, while a mixed result should be reported as a boundary condition rather than hidden behind an aggregate. The current boundary claim is [TBD].

Failure family	Evidence source required
OCR and text-in-image errors	`[TBD]`; `[TBD]`
Answer-format drift	`[TBD]`; `[TBD]`
Perception/cognition mismatch	`[TBD]`; `[TBD]`
Coverage gap after selection	`[TBD]`; `[TBD]`

Table 5: Failure-case scaffold. Examples and counts must be backfilled from saved raw evaluation rows, not from anecdotal inspection.

7 Failure Cases

The failure analysis will be drawn only from saved raw evaluation rows. Planned categories include OCR failures, multiple-choice parsing errors, perception/cognition mismatches, answer-format drift, and source-regime gaps. Representative examples are `[TBD]`.

The failure-case section is also where a negative or mixed selector outcome should be made useful. If CAQS improves OCR-sensitive rows but hurts broad recognition, or if scalar-quality controls win on one benchmark family, the final paper should state that boundary directly. The present scaffold records only the planned taxonomy and `[TBD]`.

8 Conclusion

This scaffold defines a controlled data-centric question for vision-language instruction tuning but does not answer it yet. The final conclusion must be rewritten after the full-data baseline, subset/control matrix, ablations, and benchmark evaluations produce canonical artifacts.

Limitations

The current manuscript is blocked on full-scale evidence. It does not contain completed benchmark metrics, paired significance tests, or final failure cases; the verified bibliography scaffold is present, but final citation audit remains pending until the evidence-stable paper. The planned comparison is limited to the LLaVA-1.5-7B LoRA setting and the four benchmark families named in the experiment plan unless later artifacts extend the evidence base.

Evidence limitations. The scaffold must not be read as a completed empirical paper. The full-data baseline has not yet been collected as a valid adapter and the subset/control matrix has not been launched. Any future final claim must survive the placeholder ledger, the run collection reports, the claim graph, and the format preflight. If the measured evidence rejects the data-efficiency hypothesis, the final paper should either pivot with operator approval or report the negative result under an explicitly approved scope.

Scope limitations. The design isolates one backbone family, one LoRA tuning recipe, and four benchmark families. That isolation is useful for data-centric attribution, but it limits claims about other multimodal backbones, larger parameter scales, proprietary evaluators, open-ended conversation quality, and safety-sensitive deployment settings. The planned failure taxonomy can characterize some boundary cases, but it will not substitute for broader deployment evaluation.

Ethics

The study uses public or officially released multimodal datasets and benchmark sources according to their access terms. Dataset provenance, license/access status, and filtering decisions must remain auditable in the final artifact package. The paper should avoid overclaiming benchmark performance and should report negative or mixed outcomes directly if the measured evidence does not support the hypothesis.

Data provenance and release. Because the contribution is data-centric, the final release must distinguish between source manifests, derived selection metadata, and trained adapters. It should not redistribute restricted images or annotations beyond their licenses. It should publish enough manifest hashes, filtering rules, and subset IDs for reproducibility while respecting source-dataset terms.

Benchmark interpretation. The four planned benchmark families are proxies for multimodal capability, not guarantees of real-world reliability. The paper should avoid claims about accessibility, medical, legal, or safety-critical usage. If the selected data changes model behavior in ways that increase hallucination, answer-format brittleness, or OCR error rates, those effects should be reported in the failure analysis rather than treated as incidental.

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A Reproducibility

The final appendix will list the released manifests, training configuration summaries, evaluation harness versions, benchmark splits, result-table generation commands, and STOP/resume policy. This scaffold intentionally omits machine-local execution details from rendered prose.

A.1 Artifacts To Backfill

The final reproducibility appendix should enumerate the full-data baseline adapter inventory, subset manifests, control manifests, benchmark raw rows, aggregate result tables, significance artifacts, and failure-case sample rows. In this draft those artifacts are represented by `[TBD]`, `[TBD]`, `[TBD]`, `[TBD]`, `[TBD]`, and `[TBD]`.

A.2 Re-run Boundaries

Paper-facing reproducibility should describe the framework, model family, data conditions, benchmark versions, and decoding or scoring settings. Host-specific execution details and private orchestration metadata belong in internal manifests and should not appear in the rendered paper.